Curriculum vitae

DURAFSHAN SAKEENA SYED, Ph.D.

Assistant Project Scientist
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Research Overview

Neuroscientist investigating the circuit, developmental, and computational principles that govern coordinated motor behavior. Using *Drosophila* as a genetically tractable model, my work bridges developmental neurobiology and systems neuroscience to uncover how inhibitory premotor circuits and descending pathways generate rhythmic limb movements, orchestrate behavioral transitions, and enable motor flexibility. My long-term goal is to establish a circuit-to-behavior framework for inter-limb coordination and motor adaptability, including how circuits reorganize after perturbation, providing mechanistic insight relevant to neural repair, movement disorders, and translational neuroscience.

Education

Ph.D.	National Centre for Biological Sciences, Tata Institute of Fundamental Research (NCBS-TIFR), Bangalore, India. 2017	Neuroscience
Master of Science	University of Kashmir, Srinagar, J&K, India. 2008	Biotechnology
Bachelor of Science	University of Kashmir. 2006	Biology

Appointments

Assistant Project Scientist, UCSB, Neuroscience Research Institute (NRI) — *2023–Present*
Advisor: Prof. Julie H. Simpson

- Lead independent projects uncovering neural circuit and computational principles underlying the generation and transition of sequential motor behaviors; supervise undergraduate researchers, coordinate multi-lab collaborations, and contribute to manuscripts and grant proposals.

Postdoctoral Research Fellow, UCSB, NRI, Molecular, Cellular, and Developmental Biology MCDB — *2017–2023*
Advisor: Prof. Julie H. Simpson

- Discovered how inhibitory premotor circuits orchestrate rhythmic leg movements and enable precise grooming sequences in *Drosophila.*
- Identified descending command neurons coordinating sequential grooming actions
- Developed innovative behavioral and connectomics pipelines to probe circuit-level mechanisms of sequential motor behaviors.
- Contributed data and writing to NSF CAREER Award IOS-1943276 (PI: J.H. Simpson)

Research Associate (Bridging-Postdoc), NCBS-TIFR, Bangalore, India *2017*
Advisor: Prof. K. VijayRaghavan. Collaborator: Late Prof. Heinrich Reichert

- Investigated molecular mechanisms underlying flight muscle development in *Drosophila*

Doctoral Research Scholar, NCBS-TIFR, Bangalore, India — *2009–2017*
Advisor: Prof. K. VijayRaghavan. Collaborator: Late Prof. Heinrich Reichert

- Uncovered molecular and cellular mechanisms regulating leg motor neuron architecture essential for walking.
- Characterized developmental lineage and organization of adult leg motor neurons, revealing principles of myotopic map formation.

- Supervised genetic screens for walking behavior and mentored junior researchers

Graduate rotations, NCBS-TIFR, Bangalore, India — *2009*

- *With Prof. K. VijayRaghavan*: Characterized Vglut enhancer line expression in adult leg motor neurons and interneurons.
- *With Prof. Upinder Singh Bhalla*: Performed calcium imaging in *Drosophila* olfactory receptor neurons; established the system in a mouse-focused lab.

Junior Research Fellow, NCBS-TIFR, Bangalore, India — *2008–2009*
Advisor: Late Prof. Veronica Rodrigues

- Conducted genetic screens in glia and neurons to identify candidate genes regulating behavior.
- Traced developmental origins of embryonically born adult leg motor neurons using MARCM.

M. S. Biotechnology Dissertation , Indian Institute of Integrative Medicine, Council of Scientific and Industrial Research (IIIM/RRL, CSIR), Plant Biotechnology, Jammu, India — *2007–2008*
Advisor: Dr. Rekha Sapru Dhar

- Generated a cDNA library and identified micro-satellite regions in the genome of a medicinal plant, *Withania Somnifera*

Publications

- **D. S. Syed***, P. Ravbar, J. H. Simpson*. (2025) *Inhibitory circuits control leg movements during Drosophila grooming*. *eLife* (*corresponding author)
- E Lesser, A.W. Azevedo, J.S. Phelps, L. Elabbady, A. Cook, **D. S. Syed,** B. Mark, S. Kuroda, A. Sustar, A. Moussa et al. (2024) *Synaptic architecture of leg and wing premotor control networks in Drosophila*. *Nature*
- **D. S. Syed,** S.B.M. Gowda, O.V. Reddy, H. Reichert, and K. VijayRaghavan. (2016) *Glial and neuronal Semaphorin signaling instruct the development of a functional myotopic map for Drosophila walking*. *eLife*

Book Chapter

- **D. S. Syed**, M.S. Khan, U. Afnan, T. Maqbool. (2024) *Muscular dystrophy: underlying cellular and molecular mechanisms and various nanotherapeutic approaches* *Springer Nature*

Manuscript under preparation

- **D. S. Syed***, P. Ravbar*, B. Masood, J. H. Simpson. (2026) *Structured transitions between motor attractors underlie behavioral flexibility in Drosophila grooming*. *Manuscript under preparation* (*equal contribution)
- J. H. M. Soffers, **D. S. Syed,** F. Loesche, B. Chowdhury, L. Abhilash, M. Q. Clark, M. H. Syed . (2026) *Pueblo Brain Science: A community-centered Drosophila neuroscience curriculum connecting development, connectomics, and behavior. Manuscript under preparation*

Work in progress

- **D. S. Syed**, J. H. Simpson. (2025) *A multitasking descending neuron coordinates grooming actions*. *Manuscript under preparation*

Oral presentations

- Society for Neuroscience (SfN) 2024 Nanosymposium — An inhibitory circuit for limb coordination during *Drosophila* grooming.
- National Science Foundation (NSF) IOS Virtual Colloquium (2023) — Neural control of a motor sequence in *Drosophila*.

- Pueblo Brain Science, UNM (2023) — How the nervous system controls sequential actions.
- NCBS-TIFR Seminar (2022) — Inhibitory circuits coordinate grooming behavior.
- Kavli Brain Showcase, UCSB (2018) —The role of inhibition in sequential action choice.
- Johns Hopkins University (2015) — Development of leg motor neuron architecture.
- Eighth JK Science Congress, KU, J&K (**2012**) —Semaphorins and leg motor neuron architecture.
- 2nd Indian Ocean Rim Muscle Colloquium, inStem, Bangalore (**2012**)—Developmental origins of leg motor neurons

Poster presentations

- **D. S. Syed**, P. Ravbar, J.H. Simpson (**2025**). Inhibitory premotor circuits coordinate rhythmic leg actions and behavior transitions in *Drosophila* grooming, SfN 2025, San Diego
- **D. S. Syed**, P. Ravbar, J.H. Simpson (**2025**). Inhibitory premotor circuits coordinate rhythmic leg movements, Computing Cords: Sensorimotor Control by Nerve Cord Circuits Across Species Conference, Janelia Research Campus, Virginia
- **D. S. Syed**, P. Ravbar, J.H. Simpson (**2024**). An inhibitory circuit for limb coordination during *Drosophila* grooming, Neural circuits meeting, Cold Spring Harbor Laboratory (CSHL), New York
- **D. S. Syed,** P. Ravbar, J.H. Simpson (**2023**). Inhibitory circuits coordinate leg movements, *Drosophila* Neurobiology meeting, CSHL, New York
- Y. Zhang*, **D. S. Syed**, J.H. Simpson (**2023**), Descending neurons regulate grooming actions in *Drosophila* melanogaster, *Undergraduate Research and Creative Activities (URCA) poster
- **D. S. Syed**, P. Ravbar, S. Yoshikawa, L. Guo, N. Zhang, J. S. Phelps, B. Mark, J.C. Tuthill, W.A. Lee, J.H. Simpson (**2022**), Inhibitory circuits coordinate *Drosophila* grooming, Neural circuits meeting, CSHL, New York
- **D. S. Syed**, J.H. Simpson (**2021**), The importance of inhibition in coordinating grooming actions, *Drosophila* neurobiology, CSHL, Virtual
- **D. S. Syed**, J.H. Simpson (**2021**), Inhibitory circuits regulate sequential action selection in grooming, 62nd Annual *Drosophila* Research Conference, Virtual
- J. Carranza*, **D. S. Syed**, J.H. Simpson (**2021**), Mapping the Inhibitory Circuits that Control Motor Sequence in *Drosophila*, *California Alliance for Minority Participation (CAMP), UCSB
- **D. S. Syed**, J.H. Simpson (**2021**), The importance of inhibition in grooming action choice, Society for Neuroscience (SfN) Global Connectome, Washington, Virtual
- N. Hill, P. Lawton, K. Ross, P. Tang, M. Teitz, D. Wright, **D. S. Syed,** S. Yoshikawa, J.S. Phelps, W.A. Lee, J.H. Simpson, B. Mark, J.C. Tuthill, ZETTA AI LLC. (**2021**), An undergraduate team project to reconstruct neural circuits controlling *Drosophila* grooming behavior using an electron microscopy dataset for the fly's ventral nerve cord, SfN Global Connectome, Washington
- L. Guo, **D. S. Syed**, J.H. Simpson (**2019**), The role of descending neurons in grooming behavior choice, CSHL, New York
- **D. S. Syed**, J.H. Simpson(**2019**), The role of inhibition in sequential action choice in grooming, CSHL, New York
- J.H. Simpson, N.Zhang, S. Yoshikawa, **D. S. Syed**, P. Ravbar, J. Mueller, S.Marella, L. Guo (**2019**), "Neural circuits control fly grooming over several timescales", 60th Annual *Drosophila* Research Conference, Dallas
- S. Tamir*, **D. S. Syed**, J.H. Simpson (**2019**), The roles of descending neurons in behavioral control", RACA-CON, UCSB *Undergraduate Research and Creative Activities (URCA) poster
- **D. S. Syed**, Swetha B.M., V.R. Onteddu, H. Reichert, K.VijayRaghavan (**2015**), Semaphorins instruct the development of functional motor neuron architecture in the *Drosophila* walking circuit, CSHL, New York

- **D. S. Syed**, Swetha B.M., V.R. Onteddu, V. Rodrigues, H. Reichert, K.VijayRaghavan (**2014**), Semaphorins are involved in the development of functional motor neuron connectivity in the *Drosophila* walking circuit, Behavioral Neurogenetics of larval *Drosophila*: Molecules, Circuits, Computation and Robotics meeting, University of Tokyo, Japan.
- V.R. Onteddu, T.A. Masoodi, **D. S. Syed** and K. VijayRaghavan (**2010**), Generation of neuro-genetic tools to study motor neuron development in *Drosophila*, Maggot Meeting: Neural Circuits and Behaviour, Bangalore
- **D. S. Syed,** Swetha B.M., S. Palliyil, V. Rodrigues, K. VijayRaghavan (2010), Role of Semaphorins in generation of myotopic map in *Drosophila*, Maggot Meeting: Neural Circuits and Behaviour, Bangalore

Honors & Awards

- Individualized Professional Skills Grant Program, and Neuroscience Research Institute-UCSB, Travel Grants, UC Santa Barbara, USA (2019, 2020)
- Department of Biotechnology, Government of India (DBT-GOI), International Travel Award (2015)
- Best Poster Presentation- Biology Across Scales, NCBS-TIFR (2015)
- Asia Pacific Developmental Biology Network (APDBN)— International Travel Award (2014)
- University of Tokyo — International Travel Award(2014)
- Best Oral Presentation, JK Science Congress (2012)
- Senior & Junior Research Fellowship, Tata Institute of Fundamental Research, awarded through a nationally competitive selection process(2009-2016)
- Qualified the Joint National Entrance Examination for Ph.D. Programs in Life Sciences, securing admission to NCBS-TIFR and TIFR Mumbai — a nationwide competitive exam(2009)
- Ranked All-India 2nd in the national competitive entrance examination for Ph.D. admission in science faculty, Indian Institute of Science (IISc), Bangalore, India (2009)
- Qualified UGC-NET (National Eligibility Test) in Life Sciences, Government of India — nationwide examination for eligibility to teach for lectureship (2008)
- DBT-GOI fellowship for Masters in Biotechnology (2006-2008)

Teaching Experience

- Instructor, EM Reconstruction & Connectome Analysis (MCDB 199), UCSB — 2021–2024
- Teaching Assistant, Neurophysics of Locomotion, KITP Summer Course, UCSB — 2023
- Instructor, NeuroCure23 Program, UNM — 2023
- Workshop Instructor, Pueblo Brain Science Program, UNM — 2023
- Workshop Instructor, NCBS Outreach Courses — 2010–2017

Mentoring Experience

- Mentored ~40 UCSB undergraduate researchers (2018–2025); students presented at SfN, URCA, and CAMP
- Guided trainees contributing to connectomics analyses, UCSB (2021–2024)
- Mentored junior graduate students and research interns at NCBS (2012–2017)
- Mentored undergraduate researchers completing Honors theses in MCDB, UCSB (2019, 2021, 2022)

Synergistic Activities

- Equity advisor for hiring Assistant Project Scientists, UCSB (2023)
- Peer reviewer/co-reviewer for eLife, Neuron, Nature Communications, Current Biology (2018-2024)

- Outreach: Pueblo Brain Science (UNM), NeuroCure23, Women in STEM Mentorship Program (UCSB), STEM Pals, JKScientists core member, Volunteer Teacher for underprivileged school children in Kashmir
- Contributions to unpublished data, training, and feedback for multiple scientific publications (Badsha et al, 2012, Gowda et al., 2018, Guo et al., 2022, Simpson 2024, Yoshikawa et al., 2024)